

MatrixHR

Detailed Project Specification

The Connected HR Platform Built for South Asia and the Middle East

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About this document: This is the complete reference specification for MatrixHR. It describes every module in depth, every functionality within each module, the data architecture, integration ecosystem, and the strategic scaling plan across product, infrastructure, team, and market. Use this as the source of truth during build, hiring, and fundraising conversations.

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1. Executive Summary

MatrixHR is a multi-tenant SaaS HR and payroll platform designed for businesses in Pakistan, the Gulf Cooperation Council (GCC), and the broader South Asian market. The product addresses gaps left by global platforms like BambooHR, Rippling, and HiBob, which do not support local payroll, local labor law, regional languages, or local payment infrastructure.

MatrixHR competes on three primary axes: **localization** (Pakistani and GCC payroll, FBR/EOBI/PF/gratuity, Urdu and Arabic with RTL, PKR/AED/SAR pricing), **experience** (modern UI on par with Linear and Stripe, WhatsApp-native workflows, mobile-first design), and **economics** (transparent pricing in local currency at 60–70% lower cost than equivalent global tools).

The platform consists of **14 core modules**, organized into three release phases over 18 months, with a fully scalable architecture capable of serving 100,000+ employees across thousands of tenants.

2. Product Vision & Market Positioning

Vision Statement

To become the default HR operating system for businesses in South Asia and the Middle East — combining global product quality with deep regional fluency.

Target Market Segments

- **Software houses and IT services firms** — 20–500 employees, our home market.
- **Manufacturing and industrial firms** — shift-based workforce, biometric attendance.
- **Retail chains and quick-service restaurants** — distributed locations, hourly workers.
- **Schools, hospitals, and educational institutions** — multi-role staff, structured policies.
- **Professional services** — legal firms, agencies, consultancies, accountancies.
- **Construction and field-services companies** — geo-fenced attendance, multi-site operations.

Competitive Positioning Map

Competitor	Strength	Weakness vs. MatrixHR
BambooHR	Mature, polished UI, brand	No local payroll, USD pricing, no Urdu
ResourceInn (PK)	Pakistan presence, local features	Dated UI, slow innovation, weak mobile
PeopleQlik (PK)	MENA reach, broad features	Premium pricing, complex onboarding
Bayzat (UAE)	GCC focus, modern UI	No Pakistan, premium pricing
Keka (India)	Strong product, India coverage	No Pakistan localization
Rippling	Best-in-class IT + HR	Enterprise-only, very expensive

3. Platform Architecture Overview

Architectural Style

MatrixHR is built as a **modular monolith** on the backend with strict module boundaries, allowing future extraction into microservices without rewrites. The frontend is a server-rendered Next.js application with a separate React Native mobile client. All clients communicate through a unified REST + WebSocket API.

Core Technology Stack

Layer	Technology	Purpose
Backend Framework	NestJS (Node.js 20, TypeScript)	Modular API server
Database	PostgreSQL 16	Primary transactional store
Cache & Queue	Redis 7 + BullMQ	Caching, sessions, background jobs
ORM	Prisma	Type-safe database access
Search	Meilisearch (later: OpenSearch)	Employee, document, log search
Web Frontend	Next.js 14 + Tailwind + shadcn/ui	Customer-facing web app
Mobile	React Native (Expo) + NativeWind	iOS + Android apps
File Storage	AWS S3 / Cloudflare R2	Documents, photos, payslips
Email	Resend / AWS SES	Transactional email
WhatsApp	Meta WhatsApp Business Cloud API	Notifications, approvals
SMS Fallback	Twilio / local Pakistani providers	Backup channel
Monitoring	Sentry, Grafana, Loki, Prometheus	Errors, metrics, logs
CI/CD	GitHub Actions	Automated testing & deployment
Hosting	DigitalOcean / Hetzner (Phase 1), AWS (Phase 2)	App Platform Hosting
CDN	Cloudflare	Edge caching, DDoS protection

Multi-Tenancy Model

MatrixHR uses a **single-database, shared-schema** multi-tenancy model with row-level security (RLS) and a mandatory tenant_id on every business table. Tenants are identified via subdomain routing (acme.matrixhr.com). At scale, premium tenants can be moved to dedicated database shards without code changes.

4. Detailed Module Specifications

Each of the 14 modules below is documented with its purpose, complete feature list, key entities, integrations, and unique selling points.

4.1 Authentication & Multi-Tenancy

Purpose

Manage company sign-up, user authentication, session lifecycle, role-based access, and tenant isolation across the entire platform.

Functionalities

- Company sign-up wizard (3 steps: account → company → confirmation) with email verification
- Subdomain provisioning (acme.matrixhr.com) on sign-up
- Email + password login with bcrypt-hashed credentials
- Two-factor authentication (TOTP via Google Authenticator, Authy)
- Social login: Google, Microsoft (optional)
- Single Sign-On (SSO) for enterprise: SAML 2.0, OAuth 2.0
- JWT access tokens (15 min) + rotating refresh tokens (7 days)
- Password reset via email with secure token (1-hour expiry)
- Password strength enforcement (configurable per tenant)
- Account lockout after N failed attempts (configurable)
- Five built-in roles: Super Admin, Company Admin, HR Manager, Manager, Employee
- Custom roles with granular permissions (Phase 2)
- Session management dashboard — view active sessions, force logout
- IP allowlist per tenant (enterprise feature)
- Tenant switching for users belonging to multiple companies
- Tenant suspension, deletion (soft), and reactivation workflows
- Audit log of every authentication event

Key Entities

- **Tenant:** id, name, subdomain, plan, status, timezone, currency, locale, branding
- **User:** id, tenant_id, employee_id, email, password_hash, role, status, 2fa_enabled
- **Session:** id, user_id, refresh_token_hash, device_info, ip, expires_at
- **Role:** id, tenant_id, name, permissions_json (custom roles, Phase 2)

4.2 Employee Database (HRIS)

Purpose

Central source of truth for every employee — personal, employment, financial, and document data. This is the foundation every other module depends on.

Functionalities

- Full employee profile: personal, employment, compensation, banking, documents
- Personal fields: full name, father's name, CNIC, gender, DOB, marital status, religion (optional), nationality, blood group, address (current + permanent), emergency contact, photo
- Employment fields: employee ID, designation, department, manager, employment type (permanent/contract/probation/intern), date of joining, probation end date, confirmation date, work location, shift, reporting structure
- Compensation fields: base salary, allowances breakdown, bank account, IBAN, NTN
- Document vault: CNIC scan, education certificates, experience letters, contracts, NDAs, ID cards — categorized, versioned, with expiry alerts
- Custom fields per tenant (JSONB) — text, number, date, dropdown, multi-select
- Department & sub-department hierarchy (unlimited depth)
- Designation library with grade levels
- Org chart visualization (interactive, zoomable, exportable to PDF/PNG)
- Employee directory with rich search (name, CNIC, email, phone, department)
- Advanced filters: department, status, joining date range, employment type, location, manager
- Bulk operations: CSV import wizard, bulk export, bulk status update, bulk notification
- Employee timeline: chronological history of role changes, promotions, salary revisions
- Confirmation tracking: probation employees flagged for review before confirmation date
- Birthday and work anniversary tracker with automated wishes (email/WhatsApp)
- Employee status workflows: Active → On Leave → Suspended → Terminated → Resigned
- Termination workflow: exit interview, asset return, final settlement, experience letter
- Re-hire tracking: link past records when employee returns
- Family information: spouse, children, dependents (for benefits and insurance)
- Skills inventory: tagged skills with proficiency levels, certifications, languages
- Print employee card (PDF) and employee summary

Key Entities

- **Employee:** 60+ fields covering personal, employment, financial dimensions
- **Department:** id, name, parent_id, head_employee_id, cost_center
- **Designation:** id, name, grade, department_id
- **EmployeeDocument:** id, employee_id, type, file_url, issue_date, expiry_date
- **EmployeeFamily:** id, employee_id, relation, name, dob, dependent
- **EmployeeSkill:** id, employee_id, skill, proficiency, certified
- **EmploymentHistory:** id, employee_id, event, from, to, salary_before, salary_after

4.3 Leave & Time-Off Management

Purpose

Manage every type of leave from policy definition to balance tracking, request workflows, approvals, and integration with attendance and payroll.

Functionalities

- Configurable leave policies per tenant — unlimited types
- Pre-built policy templates: Annual, Casual, Sick, Maternity (16 weeks Pakistan), Paternity, Hajj, Bereavement, Marriage, Compensatory Off, Unpaid
- Per-policy rules: applicable to whom, gender restriction, eligibility (after N months)
- Accrual rules: yearly grant, monthly accrual (1.5 days/month), pro-rata for joiners
- Carry-forward rules: unlimited, capped (e.g., max 10 days), expires after N months
- Encashment rules: at year-end or on termination
- Maximum balance cap to prevent unlimited accumulation
- Negative balance allowance (advance leave) — configurable
- Half-day leave (morning/afternoon) and quarter-day support
- Hourly leave for flexible workplaces
- Sandwich rule handling — whether weekends/holidays in between are counted
- Leave request form: dates, type, reason, attachment (medical certificate, etc.), approver preview
- Multi-level approval chains (configurable, sequential)
- Parallel approval routing (Phase 2)
- Auto-approve rules (e.g., 1-day casual leave auto-approves)
- Delegate approval when manager is on leave
- Approval via web, mobile, WhatsApp interactive buttons, email
- 'Who's Out' calendar — month/week views, filter by team/department
- Team calendar overlay — see colleagues' leaves to avoid clashes
- Public holidays preloaded for Pakistan, UAE, Saudi (current + future years)
- Hijri calendar overlay alongside Gregorian
- Religious holidays (Ramadan, Eid) auto-calculated using Hijri rules
- Ramadan working hours mode (shortened shifts, applied automatically)
- Optional holidays — employees choose which to take from a list
- Leave balance dashboard with progress bars per policy
- Leave history with timeline view, filter, export
- Year-end leave summary report
- Leave forecast — projected balance after planned leaves
- Integration with payroll: LWP (Leave Without Pay) auto-deducted from salary
- Integration with attendance: leave days excluded from absent count

- Compensatory off auto-generation when employee works on holiday
- Out-of-office auto-responder email and WhatsApp status

4.4 Attendance & Time Tracking

Purpose

Track when employees start and stop work, where they work from, and how their hours roll up into payroll and compliance reports.

Functionalities

- Web-based clock-in/out with single-click action and live current time
- Mobile clock-in/out (iOS, Android) with geo-fence and selfie verification (optional)
- Biometric integration with ZKTeco, Suprema, Anviz, Hikvision devices
- QR code-based clock-in (employee scans office QR poster)
- RFID card-based attendance for factory floors
- Manual punch entry by HR with audit trail
- Multi-device sync — employee can clock in on web, out on mobile
- Configurable shifts: fixed (9am–6pm), flexible (any 8 hours), split, rotational
- Shift assignment per employee, per department, or per role
- Rotational shift scheduler (auto-rotate weekly/biweekly)
- Shift templates and rota planning for retail/factory environments
- Grace period for late arrival (configurable, e.g., 10 minutes)
- Half-day deduction rule when late beyond threshold
- Overtime calculation: post-shift hours, holiday work, weekend work
- Overtime approval workflow (manager approves before payroll)
- Geo-fencing: clock in only when within X meters of office GPS location
- Multiple office locations support — each with its own geo-fence
- Field staff mode — clock in from anywhere with location captured
- Break tracking: lunch break, tea breaks (configurable, paid/unpaid)
- Idle time detection (Phase 2, for desk workers)
- Daily attendance log with: clock in, clock out, hours, status, location, source
- Monthly attendance heatmap (visual calendar with color-coded days)
- Absent without leave (AWOL) auto-flagging
- Regularization requests — employee explains missed punch, manager approves
- Bulk regularization for system outages
- Attendance reports: daily, weekly, monthly, by employee/department/location
- Late arrival report, early departure report, overtime report
- Export to Excel, PDF
- Real-time dashboard: who's in, who's out, who's late (HR view)

- Attendance policy enforcement: warning email after N late arrivals
- Integration with payroll: late deductions, overtime payments calculated automatically

4.5 Onboarding & Offboarding

Purpose

Standardize and automate the employee lifecycle entry and exit — ensuring consistent experience, compliance, and asset accountability.

Functionalities — Onboarding

- Workflow templates per role/department (Software Engineer onboarding, Sales onboarding, etc.)
- Pre-joining checklist: offer letter, document submission, system access requests
- Day-1 checklist: orientation, asset issuance, account creation, manager introduction
- 30/60/90-day milestones with auto-reminders
- New hire portal — employee logs in pre-joining to complete tasks
- Document collection: CNIC, education certificates, experience letters, photos
- Document verification flow (HR marks as verified)
- Asset assignment tracker: laptop, ID card, access cards, company SIM, vehicle
- Asset return on exit with condition assessment
- Auto-generated welcome email with company info, team contacts, first-day agenda
- Buddy assignment — auto-assign a peer for first week
- Welcome kit ordering integration (third-party vendors)
- IT provisioning checklist: email, Slack, GitHub, AWS, project tools
- Training assignment from LMS based on role
- Onboarding progress dashboard for HR and manager
- Onboarding feedback survey at 7, 30, 90 days
- Probation review reminders before confirmation date
- Confirmation letter auto-generation on successful probation completion

Functionalities — Offboarding

- Resignation submission workflow (employee submits, manager and HR notified)
- Notice period calculator based on employment contract
- Exit interview form with structured and open-ended questions
- Knowledge transfer checklist
- Asset return tracking with damage assessment
- Access revocation checklist (email, Slack, GitHub, VPN, building access)
- Final settlement calculation: unused leave encashment, gratuity, pending bonuses
- Experience letter auto-generation with custom template
- Reference letter generation (optional)
- Alumni network — keep ex-employees in a separate database for re-hiring
- Re-hire eligibility flag based on exit interview
- Termination workflow (involuntary) with legal compliance checklist

- Exit analytics: turnover rate, reasons for leaving, regrettable vs. non-regrettable

4.6 WhatsApp Integration

Purpose

Make MatrixHR accessible through WhatsApp — the messaging platform every Pakistani and Middle Eastern employee already uses daily. This is our killer differentiator versus global competitors.

Functionalities

- WhatsApp Business Cloud API integration (Meta-approved)
- Templated message library with multi-language support (English, Urdu, Arabic)
- Trigger-based automation engine with configurable rules
- Two-way conversational interface (chatbot)
- Interactive buttons for approvals (Approve / Reject / View Details)
- Rich media: send payslip PDFs, document attachments, photos
- Notification triggers: leave request submitted (to approver), leave decision (to applicant), payslip ready, attendance reminder, birthday/anniversary, document expiry, policy update
- Conversational commands: type 'balance' to see leave balance, 'payslip' for latest, 'attendance' for this month, 'team' to see who's out today
- Apply for leave entirely via WhatsApp (guided conversation)
- Clock in via WhatsApp (with location)
- Recruitment: receive job applications via WhatsApp
- Candidate communication: schedule interviews, share offer letters
- Broadcast announcements to all employees (one-way)
- Opt-out support for employees who prefer email only
- Audit log of every WhatsApp message sent and received
- Delivery status tracking (sent, delivered, read)
- Template approval status tracking with Meta
- Fallback to SMS if WhatsApp delivery fails
- Admin panel: configure templates, triggers, broadcast lists
- Analytics: messages sent, response rates, automation effectiveness

4.7 Payroll Engine

Purpose

Calculate, approve, and disburse employee compensation with full compliance to Pakistani and GCC tax law. The payroll engine is the most legally sensitive module — accuracy is non-negotiable.

Functionalities — Salary Structure

- Salary structure builder with drag-and-drop components
- Earning components: Basic, House Rent Allowance (HRA), Conveyance, Medical, Utility, Special Allowance, Children Education, Performance Allowance, Acting Allowance
- Per component: fixed amount, percentage of basic, or formula-based
- Multiple salary structures per tenant (e.g., for different grades or regions)
- Effective-dated salary revisions with full history
- Salary increment workflow (manager proposes, HR approves)
- Compensation benchmarking insights (Phase 3)

Functionalities — Tax Engine (Pakistan)

- FBR salary income tax slabs for the current fiscal year (auto-updated annually)
- Annual tax projection divided by 12 for monthly deduction
- Tax adjustments: medical reimbursement, donations to approved charities, Zakat
- Tax credits: investment in approved schemes, voluntary pension contributions
- Bonus tax handling: spread across remaining months OR single-month deduction (configurable)
- Mid-year salary changes recalculate tax accurately
- Year-end tax certificate generation per employee (Form 16 equivalent)
- WHT (Withholding Tax) reports for FBR submission
- Tax filing data export ready for FBR portal upload

Functionalities — Statutory Deductions (Pakistan)

- **EOBI:** 1% employee + 5% employer of minimum wage (provincial-specific)
- **Social Security (SESSI/PESSI/etc.):** Province-specific contributions
- **Provident Fund:** Configurable percentage, employee + matching employer contribution
- **Gratuity:** One month's basic salary per completed year of service (auto-calculated on termination)
- **Workers Welfare Fund (WWF):** Where applicable
- **Employees Old-Age Benefits:** Tracking of contributions for retirement

Functionalities — GCC Compliance (Phase 2)

- UAE: WPS (Wage Protection System) file generation, end-of-service gratuity, no income tax
- Saudi Arabia: GOSI contributions, end-of-service benefit, Saudization (Nitaqat) reports
- Qatar, Bahrain, Kuwait: country-specific rules
- Multi-country payroll runs from single tenant

Functionalities — Payroll Run

- Payroll cycle: monthly, bi-weekly, weekly (configurable per tenant)
- Pre-payroll checklist: attendance finalized, leaves approved, adjustments entered
- Step-by-step payroll wizard: Select Period → Review → Adjustments → Approve → Process → Distribute
- One-time adjustments: bonuses, deductions, reimbursements, advances
- Recurring deductions: loans, salary advances with auto-repayment
- Loan management with EMI calculation, interest (if any), prepayment
- Salary advance workflow (employee requests, approved, deducted next payroll)
- Payroll preview before final approval — see exact net pay per employee
- Variance report: changes from previous month flagged
- Multi-step approval: HR finalizes → Finance approves → CEO sign-off (configurable)
- Payroll lock once approved — no edits without reversal trail
- Off-cycle payroll runs for late joiners, terminations, corrections
- Year-to-date (YTD) earnings and deductions tracked
- Payroll re-run / reversal with full audit trail

Functionalities — Payslips & Banking

- Professional payslip PDF with company branding
- Bilingual payslips (English + Urdu / English + Arabic)
- Detailed breakdown: earnings, deductions, employer contributions, net pay, YTD totals
- Digital signature on payslips
- Payslip distribution: email, WhatsApp, self-service portal
- Bank file generation for direct salary disbursement:
 - Meezan Bank format
 - HBL format
 - UBL format
 - MCB format
 - Bank Alfalah, Standard Chartered, Allied Bank formats
 - UAE banks: ENBD, FAB, ADCB (WPS format)
- Cash payment option with receipt acknowledgment
- Cheque payment option with cheque number tracking
- Salary disbursement status tracking per employee

Functionalities — Reports & Compliance

- Monthly payroll summary report
- Salary register (full breakdown per employee)
- Department-wise cost analysis

- Tax deducted at source (TDS) report
- EOBI contribution report
- Provident Fund statement
- Annual tax certificate per employee (Pakistani format)
- Gratuity liability report for accounting
- Loan and advance outstanding report
- Bonus payment register
- Payroll journal entries for accounting software export (QuickBooks, SAP, Tally, Xero)

4.8 Recruitment / Applicant Tracking System (ATS)

Purpose

Streamline hiring from job opening to onboarding — with a focus on local job boards, WhatsApp candidates, and AI-assisted screening.

Functionalities

- Job requisition workflow: hiring manager raises requisition, HR approves
- Job posting with rich editor (description, responsibilities, requirements, perks)
- Multi-channel posting: company career page, LinkedIn, Rozee.pk, Mustakbil, Bayt (UAE)
- Embeddable career page widget for company website
- Custom application forms per job (add questions, screening criteria)
- Candidate sources tracking (referral, LinkedIn, job board, WhatsApp, direct)
- Employee referral program with reward tracking
- Candidate database with full history, resume search, tags
- Kanban-style hiring pipeline: Applied → Screening → Shortlisted → Interview → Assessment → Offer → Hired / Rejected
- Customizable pipeline stages per job
- AI-powered resume parsing (extract name, contact, experience, skills, education)
- AI-powered candidate ranking against job description
- Bulk candidate actions: shortlist, reject, move stage, send email
- Interview scheduling with calendar integration (Google Calendar, Outlook)
- Interview feedback forms with structured scoring
- Panel interview support (multiple interviewers, aggregated scores)
- Video interview integration (Zoom, Google Meet auto-created)
- Assessment integration (HackerRank, Codility for tech, custom forms)
- Background verification request to NADRA Verisys
- Education verification via partner services
- Reference check automation: system emails references, collects responses
- Offer letter builder with templates and e-signature
- Salary negotiation tracking
- Offer acceptance/rejection tracking
- Auto-conversion of hired candidate to employee record
- Onboarding workflow triggered on candidate acceptance
- WhatsApp candidate communication for entire pipeline
- Recruitment analytics: source effectiveness, time-to-hire, cost-per-hire, offer acceptance rate, drop-off rate per stage
- Diversity hiring tracking (optional)
- GDPR-compliant candidate data deletion after retention period

4.9 Performance Management

Purpose

Drive employee growth through continuous goal-setting, feedback, and structured review cycles.

Functionalities

- Goal-setting framework (SMART goals or OKRs)
- Cascading goals: company → department → team → individual
- Goal progress tracking with check-ins and updates
- Goal weights and priority levels
- Quarterly, half-yearly, annual review cycles (configurable)
- Customizable review templates per role/department
- Self-assessment forms
- Manager review forms
- 360-degree feedback: peers, reports, cross-functional
- Anonymous feedback option (peer reviews)
- Rating scales: numeric, descriptive, 9-box matrix
- Calibration sessions for managers to align ratings
- Performance Improvement Plans (PIPs) for underperformers
- Continuous feedback: give kudos, request feedback anytime
- Recognition wall (peer-to-peer praise feed)
- One-on-one meeting notes between manager and employee
- Career development plans with milestones
- Skill gap analysis vs. role requirements
- Succession planning matrix (Phase 3)
- Performance dashboards for HR, managers, executives
- Linkage with compensation: ratings drive salary revisions and bonuses
- Promotion workflow with approval chain

4.10 Learning Management System (LMS)

Purpose

Enable continuous learning, role-based training, and certification tracking — reducing reliance on third-party training tools.

Functionalities

- Course builder: text, video, PDF, quizzes, assignments
- Course library categorized by topic, role, department
- Pre-built compliance courses: sexual harassment, data privacy, code of conduct
- External course library integration: Coursera, Udemy, LinkedIn Learning
- Assignment of courses to employees (individual, role, department, all)
- Mandatory vs. optional course flagging
- Course deadlines with reminders
- Quiz engine with multiple question types (MCQ, true/false, short answer)
- Pass/fail thresholds with retake rules
- Certificate auto-generation on completion
- Skill tagging — completing courses adds skills to employee profile
- Progress tracking per employee, per course, per department
- Mobile learning support (offline content download)
- Onboarding-linked training (auto-assign 'Welcome to MatrixHR' to new hires)
- Compliance training expiry tracking (annual recertification reminders)
- Learning analytics: completion rates, average scores, engagement metrics
- Discussion forums per course (Phase 3)
- Live webinar integration (Phase 3)

4.11 Employee Self-Service Portal

Purpose

Empower employees to manage their own data and requests — drastically reducing HR's administrative burden.

Functionalities

- Personal dashboard: leave balance, next holiday, payslip access, pending tasks
- Profile management: update phone, address, emergency contact, family info
- Photo upload
- Document upload (with HR verification)
- Bank account update (with HR approval)
- Leave application and history
- Attendance log and regularization requests

- Payslip download (current + historical)
- Tax certificate download
- Loan and advance applications
- Expense and reimbursement claims with receipt upload
- Document requests: salary certificate, employment letter, experience letter, NOC
- Auto-generated documents within minutes (templated with employee data)
- Team directory with search and contact info
- Org chart access
- Holiday calendar
- Announcements feed
- Internal job board
- Training enrollments
- Feedback submission to HR
- Anonymous suggestion box
- Birthday and anniversary feed

4.12 Reporting & Analytics

Purpose

Transform HR data into strategic insights for managers, HR, finance, and leadership.

Functionalities

- Pre-built reports library (50+ reports):
 - Headcount reports (by department, location, gender, age, tenure)
 - Turnover and attrition reports
 - Hiring funnel and source effectiveness
 - Attendance and absenteeism trends
 - Leave consumption patterns
 - Payroll cost analysis
 - Compensation distribution and pay equity
 - Performance rating distribution
 - Training completion and skill gaps
- Custom report builder (drag-and-drop fields, filters, grouping)
- Scheduled reports: auto-email reports daily/weekly/monthly to stakeholders
- Executive dashboards: high-level KPIs, trends, alerts
- Department-level dashboards for line managers
- Predictive analytics (Phase 3): attrition risk scoring, hiring needs forecast
- Export formats: Excel, PDF, CSV

- API access for custom BI tools (Power BI, Tableau, Metabase)
- Data warehouse export for advanced analytics
- Anonymized benchmark reports (compare with industry peers, Phase 3)

4.13 AI-Powered Features

Purpose

Use large language models to automate routine HR tasks, surface insights, and create a conversational HR assistant.

Functionalities

- **Ask MatrixHR:** conversational AI assistant for employees and HR
 - 'What's my leave balance?'
 - 'How do I apply for maternity leave?'
 - 'What's our work-from-home policy?'
- **Resume Screening:** AI ranks candidates against job description, highlights gaps
- **Job Description Generator:** AI drafts job descriptions from role title and key requirements
- **Interview Question Generator:** AI creates tailored interview questions
- **Performance Review Summarizer:** AI condenses self/manager/peer feedback into themes
- **Policy Assistant:** employees ask policy questions, AI answers from company handbook
- **Document Generator:** AI drafts offer letters, experience letters, NOCs
- **Sentiment Analysis:** on engagement surveys, exit interviews, feedback
- **Attrition Risk Prediction:** ML model flags at-risk employees with reasons
- **Salary Recommendation:** AI suggests salary ranges based on role, experience, market
- **Bias Detection:** review job descriptions and feedback for gender/ethnic bias
- **Email/Message Draft Assist:** HR drafts announcements with AI
- **Smart Search:** natural language search across employees, documents, policies
- Privacy-preserving: all AI calls scoped to tenant data; no cross-tenant leakage
- Configurable AI: tenants can opt out, choose preferred model, or self-host

4.14 Marketplace & Integrations

Purpose

Connect MatrixHR to the broader ecosystem of tools and services that employees and businesses use — creating revenue through partner commissions.

Productivity Integrations

- Slack: notifications, leave bot, attendance bot, announcements
- Microsoft Teams: same as Slack
- Google Workspace: SSO, calendar, drive document sync
- Microsoft 365: SSO, calendar, OneDrive
- Zoom and Google Meet: auto-create meetings for interviews and 1-on-1s
- Jira and Asana: link tasks to projects and time tracking
- GitHub and GitLab: link developer activity to performance data

- Trello and ClickUp: project linking

Accounting & Finance

- QuickBooks: payroll journal entries export
- Xero: same as QuickBooks
- SAP Business One: enterprise integration
- Tally: common in Pakistani SMBs
- Sage: international accounting

Local Pakistan/MENA Integrations

- **NADRA Verisys**: CNIC verification for employees and candidates
- **FBR IRIS**: tax filing data submission (where API available)
- **Rozee.pk and Mustakbil**: job posting and candidate import
- **SadaPay, Easypaisa, JazzCash**: salary advance and wallet integration
- **Meezan Roshan Digital, HBL, UBL, MCB APIs**: salary disbursement
- **Bayt (UAE)**: job posting in GCC market

Employee Lifestyle Marketplace

- **Foodpanda for Business**: office meal ordering, lunch credits
- **Careem Business**: corporate ride accounts
- **Sehat Kahani, Marham, oladoc**: telemedicine consultations
- **Adamjee, EFU, Jubilee**: health insurance enrollment and claims
- **Dawaai**: pharmacy discounts and medicine delivery
- **Coursera, Udemy, LinkedIn Learning**: external course access
- **Local gyms (Shapes, Structure)**: fitness memberships
- Revenue model: partner commissions (5–20%) on referred transactions

Hardware Integrations

- ZKTeco biometric devices (K40, K50, F18, F22, MB360)
- Suprema biometric devices
- Anviz and Hikvision biometric devices
- Door access control systems (HID, Suprema BioStar)
- RFID card readers
- CCTV integration for attendance verification (Phase 3)

5. Cross-Cutting Functionalities

These features apply across every module and are foundational to the platform.

Notifications System

- Multi-channel: in-app, email, WhatsApp, SMS, push (mobile)
- Per-user preferences for each channel and event type
- Notification center with read/unread, archive, snooze
- Digest mode (daily/weekly summary instead of real-time)
- Quiet hours (no notifications during configured times)
- Mention support (@employee in comments)

Workflow Engine

- Configurable approval workflows for any module
- Sequential, parallel, or conditional routing
- Delegation when approver is on leave or unavailable
- Escalation rules (auto-escalate after N hours of inaction)
- Workflow visualization with current step indicator
- Workflow templates per process type

Audit Logging

- Every data change logged: who, what, when, before, after, IP, device
- Immutable audit trail
- Searchable and exportable for compliance audits
- Per-entity history view (e.g., see all changes on an employee's record)
- Data access logs (who viewed sensitive data)

Internationalization (i18n)

- English, Urdu, Arabic — all UI strings translated
- Right-to-left (RTL) layout for Urdu and Arabic
- Date formats per locale (DD/MM/YYYY for Pakistan, MM/DD/YYYY for US)
- Number and currency formats per locale
- Hijri calendar support alongside Gregorian
- Timezone-aware (store UTC, display per user's timezone)
- Future: Bengali, Pashto, Sindhi, Punjabi support

Branding & White-Labeling

- Custom logo per tenant

- Custom primary color (auto-generates palette)
- Custom domain (Phase 2): hr.acmecorp.com instead of acme.matrixhr.com
- Branded emails with custom templates
- Branded login page
- Custom payslip templates
- Reseller mode (Phase 3): partners can white-label MatrixHR entirely

6. Mobile Application

MatrixHR Mobile is a first-class citizen — not an afterthought. Built in React Native with full feature coverage for employees and managers.

Employee Mobile Features

- Clock in/out with geo-fence and selfie verification
- Apply for leave with date picker and approver preview
- View leave balance and history
- View attendance log and request regularization
- Download and view payslips
- Submit expense claims with receipt photo capture
- View team calendar and 'Who's Out'
- Receive and respond to performance feedback requests
- Complete training courses on the go
- View and update personal profile
- Receive push notifications for all events
- Offline mode: clock in/out, view profile (syncs when online)

Manager Mobile Features

- Approve/reject leave with one tap
- Approve attendance regularization
- Approve expense claims
- View team dashboard: who's in, who's out, who's late
- Send recognition and feedback to team members
- Schedule 1-on-1s
- View team performance metrics
- Approve overtime

HR Mobile Features

- Add new employees on the go

- View company-wide attendance
- Approve high-level requests
- Send broadcast announcements
- View key HR metrics

7. Security, Compliance & Audit

MatrixHR handles sensitive personal, financial, and health data. Security is treated as a first-class engineering concern, not an add-on.

Application Security

- OWASP Top 10 protections built-in (SQL injection, XSS, CSRF, etc.)
- Content Security Policy (CSP) headers
- Rate limiting on all public endpoints
- Brute-force protection on login
- Encryption at rest: AES-256 for database and file storage
- Encryption in transit: TLS 1.3 everywhere
- Secrets management: Doppler, AWS Secrets Manager, or HashiCorp Vault
- No secrets in code or environment files in repo
- Regular dependency vulnerability scans (Snyk, Dependabot)
- Static application security testing (SAST) in CI
- Penetration testing annually by third-party
- Bug bounty program (Phase 2)

Data Privacy & Compliance

- Tenant data isolation enforced at database row level
- Right to be forgotten: full data deletion on request
- Data export per employee (GDPR-style)
- Configurable data retention policies per tenant
- Audit trail of all data access
- Anonymization of analytics data
- Compliance roadmap: GDPR, Pakistan PECA, UAE PDPL, Saudi PDPL, SOC 2 Type II
- Data residency options (Phase 3): EU, GCC, Asia data centers

Infrastructure Security

- DDoS protection via Cloudflare
- Web Application Firewall (WAF) at edge
- Private database access (no public exposure)
- VPC isolation in production
- Automated backups: daily full + hourly incremental, 30-day retention
- Point-in-time recovery (PITR) for database
- Disaster recovery plan with documented RTO/RPO
- Multi-region failover (Phase 3)

Access Control

- Role-based access control (RBAC) with 5 default roles + custom roles
- Granular permissions per module and action
- Field-level access control for sensitive data (salaries, etc.)
- Department-scoped access for managers
- Audit logs of permission changes
- Mandatory 2FA for admin roles (Phase 2)
- Session timeout and forced re-authentication for sensitive operations

8. Scaling Strategy

MatrixHR is designed to scale from 1 tenant with 10 employees to 10,000 tenants with 1 million employees — across product, infrastructure, operations, team, geography, and revenue.

8.1 Technical Scaling

Horizontal Application Scaling

- Stateless API servers — scale by adding instances behind a load balancer
- Auto-scaling based on CPU, memory, and request queue depth
- Container orchestration via Docker → Kubernetes (Phase 2)
- Sticky sessions only where necessary (none in MVP)
- Background jobs processed by separate worker pool
- Job priority queues (urgent vs. batch)

Database Scaling

- Vertical scaling first (bigger instances) — sufficient up to ~1000 tenants
- Read replicas for analytics and reporting workloads
- Connection pooling via PgBouncer
- Query optimization: indexed all `tenant_id` columns, common filter combinations
- Caching layer (Redis) for hot data: employee directory, leave policies, holidays
- Database sharding by `tenant_id` (Phase 3) when single instance bottlenecks
- Dedicated database instances for enterprise tenants (premium tier)
- Archive cold data (3+ year old logs) to cheaper storage

File Storage Scaling

- Object storage (S3/R2) scales infinitely — no concern
- CDN-cached static assets and public documents
- Per-tenant storage quotas to prevent abuse
- Lifecycle policies: move old documents to cheaper storage tiers

Caching Strategy

- Redis for session data, rate limiting, frequently-read entities
- Cache-aside pattern with TTL
- Cache invalidation on data writes
- HTTP cache headers for static and semi-static API responses
- CDN edge caching for public assets

Background Job Scaling

- BullMQ on Redis for job orchestration
- Separate queues per job type (payroll, notifications, reports)
- Worker auto-scaling based on queue depth
- Heavy jobs (payroll runs) processed by dedicated worker pool
- Job retry with exponential backoff
- Dead letter queue for failed jobs

Performance Targets

- API response time: p95 < 300ms, p99 < 800ms
- Page load: First Contentful Paint < 1.5s
- Payroll run: 1000 employees processed in < 60 seconds
- Report generation: 10,000 rows in < 10 seconds
- Search: results in < 200ms for 1M-record dataset
- Uptime SLA: 99.9% (less than 9 hours downtime/year)

8.2 Operational Scaling

Customer Onboarding

- Self-service sign-up for small tenants (<50 employees)
- Guided onboarding wizard with sample data and templates
- Onboarding videos and step-by-step guides
- White-glove onboarding for tenants with 100+ employees (paid service)
- Data migration toolkit: import from Excel, BambooHR, ResourceInn
- Dedicated onboarding manager for enterprise clients

Customer Support

- **Tier 1 — Self-service:** in-app help center, video tutorials, AI chatbot
- **Tier 2 — Email support:** 24-hour response SLA for all paying tenants
- **Tier 3 — WhatsApp/Phone:** for premium tiers, same-day response
- **Tier 4 — Dedicated Customer Success Manager:** for enterprise tier
- Local-language support (English, Urdu, Arabic) during business hours
- Knowledge base with searchable articles
- Community forum (Phase 2)
- Status page (status.matrixhr.com) for incident communication

DevOps & Release Management

- Continuous deployment with automated testing
- Feature flags for gradual rollout (LaunchDarkly or open-source equivalent)
- Blue-green deployments for zero-downtime releases
- Database migrations versioned and tested
- Rollback procedures documented and rehearsed
- On-call rotation with PagerDuty
- Incident response runbook
- Weekly release cadence (Phase 1) → daily (Phase 3)

8.3 Market & Geographic Scaling

Phase 1 — Pakistan (Months 1–12)

- Karachi-first: founders' network, software houses
- Expand to Lahore and Islamabad in months 6–9
- Target: 50 paying tenants by end of Year 1
- Localizations: PKR, English + Urdu, FBR/EOBI/PF, Pakistani banks

Phase 2 — GCC Expansion (Year 2)

- UAE first (largest market, English-friendly): Dubai, Abu Dhabi, Sharjah
- Saudi Arabia second (largest population): Riyadh, Jeddah

- Localizations: AED/SAR, English + Arabic + RTL, WPS files, GOSI, end-of-service gratuity
- Local partnerships with HR consultants and accountants
- Local sales hire in Dubai by Q2 of Year 2
- Target: 200 paying tenants by end of Year 2

Phase 3 — Broader South Asia & MENA (Year 3+)

- Bangladesh: large SMB market, similar to Pakistan
- Qatar, Bahrain, Kuwait, Oman: smaller GCC markets
- Egypt: largest Arab market by population
- India: potentially via partner channel (competitive market)
- Localizations: BDT, EGP, country-specific labor laws
- Target: 1000+ paying tenants by end of Year 3

Phase 4 — Global (Year 4+)

- English-speaking SMBs in Southeast Asia, Africa, parts of Latin America
- Partnerships with payroll providers in each new geography
- Selective expansion based on partner availability

8.4 Team Scaling

Stage	Team Size	Roles
Pre-MVP (Months 1–3)	2–3	1–2 Founders + AI-assisted dev (Antigravity)
MVP Build (Months 4–6)	3–4	Add 1 Backend Engineer, 1 QA/Support
First Customers (Months 7–12)	12–18	Add Sales, Customer Success, Mobile Dev
Year 2 — Growth	15–20	Sales team, Engineering pods, Marketing, Design
Year 3 — Scale	40–60	Multiple product pods, regional sales, ops, finance
Year 4+ — Established	100+	Full functional teams, regional offices

Engineering Team Structure (Year 2+)

- **Platform Pod:** auth, tenancy, infrastructure, core APIs
- **People Pod:** employee database, onboarding, performance
- **Time Pod:** leave, attendance, scheduling
- **Money Pod:** payroll, tax, banking integrations
- **Growth Pod:** recruitment, integrations, marketplace
- **Mobile Pod:** iOS and Android apps
- **AI Pod:** AI features, ML models, data science
- **DevOps/SRE:** infrastructure, monitoring, security

8.5 Revenue Scaling

Revenue Streams

- **Subscription revenue:** primary, per-employee-per-month (PEPM)
- **Implementation fees:** one-time setup for larger clients
- **Premium support:** dedicated CSM, priority response (enterprise)
- **Marketplace commissions:** revenue from partners (Foodpanda, insurance, etc.)
- **Custom development:** bespoke features for enterprise clients
- **Training and certification:** partner certification programs (Phase 3)
- **API access:** charge for high-volume API usage
- **White-label licensing:** resellers and partners (Phase 3)

Revenue Targets

Year	Tenants	Avg. Employees/Tenant	ARPU/month (PKR)	MRR Target (PKR)
Year 1	10–50	50	20,000–60,000	5–10 lakh
Year 2	150–200	75	30,000–90,000	30–50 lakh
Year 3	500–1,000	100	40,000–120,000	1–2 crore
Year 5	5,000+	120	50,000+	10+ crore

Unit Economics Targets

- Customer Acquisition Cost (CAC): < 3 months of revenue
- Customer Lifetime Value (LTV): 36+ months retention target
- LTV/CAC ratio: > 3x (industry standard)
- Gross margin: > 75% (typical for SaaS)
- Net Revenue Retention (NRR): > 110% (upsells + expansion)
- Churn rate: < 2% monthly (< 24% annual)

9. Pricing & Monetization

Pricing must be transparent, simple, and below global competitors while leaving room for value-add upsells.

Pricing Tiers (Initial Proposal — Pakistan)

Plan	Per Employee/Month (PKR)	Includes
Starter	250	Core HR, Leave, Attendance, Self-Service, Mobile
Growth	500	Starter + Payroll, Recruitment, Onboarding, WhatsApp
Pro	850	Growth + Performance, LMS, Advanced Reporting, API

Enterprise	Custom	Pro + AI, SSO, Dedicated CSM, SLA, Custom integrations
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Pricing for GCC Market

Plan	Per Employee/Month (AED)	Per Employee/Month (SAR)
Starter	15	15
Growth	30	30
Pro	50	50
Enterprise	Custom	Custom

Pricing Principles

- Transparent: published pricing on website (no 'request a quote' for SMB tiers)
- Annual discount: 15% off for annual prepayment
- Minimum company size: no minimum for self-serve
- Volume discounts: above 250 employees
- 30-day free trial — no credit card required
- Money-back guarantee: 30 days

10. Roadmap

Quarter	Milestones
Q1 (Months 1–3)	Architecture, Auth, Employee DB, Dashboard, App Shell
Q2 (Months 4–6)	Leave, Attendance, Onboarding, WhatsApp, MVP launch with 3 pilot clients
Q3 (Months 7–9)	Payroll engine, Pakistani tax compliance, bank file generation, 10 paying clients
Q4 (Months 10–12)	Recruitment/ATS, Performance Management, Self-Service Portal, 30+ clients
Q5 (Months 13–15)	LMS, AI features (Ask MatrixHR, resume screening), Mobile feature parity
Q6 (Months 16–18)	Marketplace integrations, advanced analytics, UAE expansion prep
Year 2 H1	UAE launch, Arabic + RTL, WPS, GOSI; 100+ tenants
Year 2 H2	Saudi launch, enterprise features (SSO, audit, SLA); 200+ tenants
Year 3	Bangladesh, Qatar, Egypt; AI-first features; 1000+ tenants

11. Success Metrics & KPIs

Product KPIs

- Daily Active Users / Monthly Active Users (DAU/MAU ratio > 50%)
- Time to first value: new tenant first payroll run within 14 days
- Feature adoption: % of tenants using each module
- Mobile app rating: > 4.5 on App Store and Play Store
- NPS (Net Promoter Score): > 40 (excellent for B2B SaaS)
- Support ticket volume per 100 tenants per month: trending down

Business KPIs

- Monthly Recurring Revenue (MRR) growth: > 15% month-over-month in Year 1
- Annual Recurring Revenue (ARR)
- Logo retention: > 90% annual
- Net Revenue Retention: > 110%
- Gross margin: > 75%
- Burn multiple: < 2x in growth phase
- Months of runway: > 18 at all times

Technical KPIs

- Uptime: > 99.9% (less than 9 hours downtime/year)
- API p95 latency: < 300ms

- Error rate: < 0.1% of requests
- Mean Time to Recovery (MTTR): < 30 minutes
- Deployment frequency: weekly (Phase 1) → daily (Phase 3)
- Code coverage: > 70% on business logic

12. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation
Local competitor undercuts price	High	Medium	Differentiate on UX, WhatsApp, AI — not price race
Tax law changes (FBR slabs)	Annual	High	Configurable tax engine, annual update process
WhatsApp policy changes (Media)	Medium	High	SMS fallback, email backup, diversify channels
PKR currency volatility	Ongoing	Medium	Price in local currency, hedge USD costs
Slow B2B sales cycles	Certain	High	Free pilots, founder-led sales, referral incentives
Engineering complexity (payments)	Medium	Critical	Comprehensive test coverage, staged rollout, audit logs
Data breach	Low	Critical	Security-first engineering, encryption, audits, insurance
Founder burnout	Medium	Critical	Realistic timelines, hire early, delegate, mental health support
Cash runway shortage	Medium	Critical	Conservative spending, focus on revenue, raise before needed
Difficult investor terms	Medium	High	Bootstrap as long as possible, optionality

This document is a living specification and will be updated as the product, market, and team evolve. Version 1.0 — June 2026.

MatrixHR — Built in Pakistan, for the world.